

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD2103 OPERATING SYSTEMS

THURSDAY, 18 MAY 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD2103 OPERATING SYSTEMS**Section A: Answer ALL questions.**

Q1. (a) Draw and label the diagram of 7-State process model. (7 marks)

(b) Referring to the following scenarios, show each state transition of Process A based on the 7-state model:

(i) Process A is created and then waits in the ready queue. (1 mark)

(ii) Due to insufficient system memory, Process A is swapped to virtual memory. (1 mark)

(iii) After the system regains sufficient memory, Process A is swapped back to the RAM. (1 mark)

(iv) Process A is selected to execute. It then requests to read the `input.txt` file stored in the hard disk drive. (1 mark)

(v) The `input.txt` file has been read. (1 mark)

(vi) Process A then continue its execution until completion. (1 mark)

(c) Table 1.1, 1.2 and 1.3 show excerpts of the Process Control Block for Process B, Process C, and Process D, respectively. Process ID 9000 is a terminal.

Table 1.1

process ID	9002
parent ID	9000
user ID	1001
state	Ready

Table 1.2

process ID	9003
parent ID	9002
user ID	1001
state	Running

Table 1.3

process ID	9004
parent ID	9000
user ID	1001
state	Blocked

Illustrate the parent-child relationship among the shell terminal, Process B, Process C, and Process D using a simple diagram. Include details such as arrows, process names and process IDs in your diagram. (8 marks)

(d) For each of the following, determine the type of real-time task (hard or soft real-time) in the application or system and briefly justify your answer:

(i) A biped robot (i.e. two-legged) calculates data read from multiple sensors and then determines its motors movement so that it will maintain its balance during walking. (2 marks)

(ii) The main task of thermal camera systems in airport is to detect travellers with fever during the Covid-19 pandemic. (2 marks)

[Total : 25 marks]

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- Q2. (a) This question is about the Completely Fair Scheduler (CFS) in the Linux kernel. Assume that the system has a uni-processor with a Red-Black tree. There are three CPU-bound processes in Ready state; A, B and C with the parameters shown in Table 2:

Table 2: Parameters for Process A, B and C

Parameters	Process A	Process B	Process C
Virtual runtime (nanoseconds)	1,483,829	1,412,123	1,552,088
Total time quantum allocated (milliseconds) per period	0.4117	1.5501	X
Nice value	Y	0	-11

The minimum virtual runtime of the red-black tree is 1,412,123 nanoseconds, and the period is set at 20 milliseconds. Using the guide provided in the Appendix A (page 9), answer the following questions:

- Calculate the values of X and Y in Table 2. (6 marks)
 - Which process in Table 2 will be selected to execute? Briefly provide your reason. (2 marks)
 - What are the virtual runtimes of Process A, Process B and Process C after the next period, respectively? (6 marks)
- (b) Assume that your main memory has a total of 6 frames. Process A and Process B are each initially allocated frames as shown in Figure 2, where frame #4 and frame #5 represent empty blocks.

frame #0	A 2
frame #1	B 0
frame #2	B 4
frame #3	A 1
frame #4	
frame #5	

Figure 2: Current state of memory frames

Assume the sequence of page requests arrival is A2, B4, A1, B0. Then, the following page trace occurs (process name and its respective page number):

A3, B2, B4, B6, B7, A3, B5, A1, A0, B4

Variable Allocation and **Global Replacement Scope** are used. Show the steps of each page replacement performed if:

UCCD2103 OPERATING SYSTEMS**Q2. (b) (Continued)**

- (i) First-In-First-Out policy is used. (5 marks)
- (ii) Clock replacement policy is used. Indicate the use-bit in your answer. Assume the use-bit in frames #0 and #3 in Figure 2 are set to 1 and frames #1, #2, #4, and #5 are set to 0. The clock arrow is now pointing at frame #4. (6 marks)

[Total : 25 marks]

- Q3. (a) You are tasked to choose a filesystem where the main goal is to store video files which are purely sequential in nature. Suggest a suitable file allocation method to be used and briefly provide the reason of your choice. (4 marks)
- (b) Consider a disk head is **now** positioned at track 200. The requests of track are in the following sequence:

211, 585, 335, 190, 135, 265, 295, 270

Answer the following questions:

- (i) Draw the seek pattern on a graph using the Shortest Service Time First (SSTF) disk scheduling algorithm starting from the track 200. Let the y-axis and x-axis represents the track number and request number, respectively. (8 marks)
- (ii) The SSTF always services the request nearest to the current head position. Does SSTF always yield the optimum total amount of head movement? Briefly justify your answer. (4 marks)
- (c) Assume the **Indexed Allocation with Fixed Size Portions** is used. Table 3 shows the content of **Index Block 4**.

Table 3: Content of Index Block 4

Block No.
10
1
14
6

- (i) Draw the allocation of File A in a disk. Show the Index Block too. A sample template of an empty disk is given in Figure 3. You just need to show the allocation from Block 0 until Block 14.

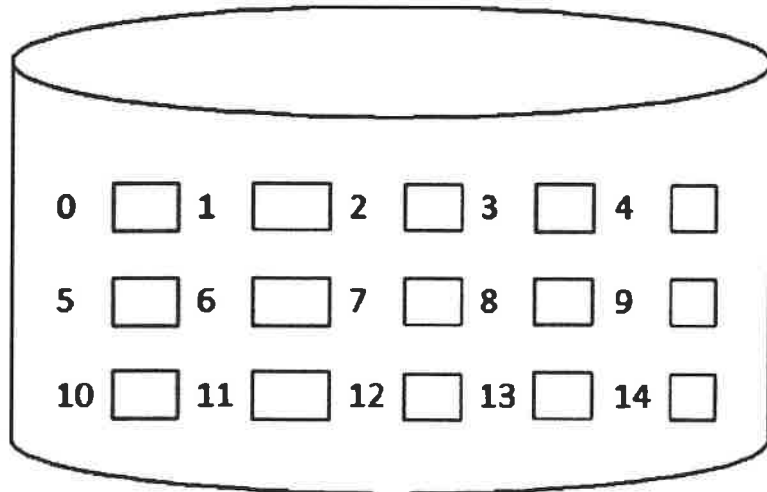
UCCD2103 OPERATING SYSTEMS**Q3. (c) (Continued)**

Figure 3: A disk with empty blocks from 0 to 14

(4 marks)

- (ii) What is the amount of memory (in MB) required to store the block bitmap? Assume the hard disk size is 2 terabytes and the file system block size is 512 bytes. (3 marks)
- (iii) Does file consolidation reduce the size of the index for the file allocation method used in this question? (2 marks)

[Total : 25 marks]

UCCD2103 OPERATING SYSTEMS**Section B: Answer ANY ONE (1) question.**

- Q4. (a) Table 4.1 shows the current allocation of resources to each process and the resources that are still needed by each process to proceed and complete their execution.

The resources currently available are $R1 = 0$, $R2 = 0$, $R3 = 1$ and $R4 = 1$. It is obvious that all the processes are deadlocked. Resolve the deadlock by *successively abort deadlocked processes with the least total resources allocated*. Show the resources available after each abortion in your workings. (9 marks)

Table 4.1

Process	Allocated				Needed			
	R1	R2	R3	R4	R1	R2	R3	R4
A	2	1	1	3	3	3	6	6
B	0	1	1	0	3	2	2	4
C	0	0	1	2	2	2	2	2
D	2	1	1	1	1	3	2	3
E	3	1	2	2	2	1	4	3

- (b) Given a user process of 8GB, page frame size is 8KB and each page table entry is 2 bytes, calculate the size of root page table in two-level scheme. Show all your calculations. (4 marks)
- (c) A 16-bit virtual address format is shown in Table 4.2 and a portion of a segment table is shown in Table 4.3. Translate the following virtual addresses to their corresponding physical address. Show all the relevant values in your answer.

Table 4.2: 16-bit virtual address format

6 bits	10 bits
Segment number	Offset

Table 4.3: Portion of a segment table

Segment number	Base	Length (decimal)
37	0000 0000 1001 1101	256
38	0000 0010 0110 0001	256
39	0000 0100 0010 0110	512

- (i) 1001100010010111 (3 marks)
- (ii) 1001111000001001 (3 marks)
- (d) **Pre-paging** is one of the fetch policies. Do you agree that pre-paging fetch policy follows the principle of locality? Justify your answer. (6 marks)
- [Total : 25 marks]

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Q5. (a) Given that a system has 1MB of memory available, apply the Buddy system allocation to the following:

(i) Memory requests in the sequence shown in Table 5.1. (5 marks)

Table 5.1

Process	Request
A	110 KB
B	60 KB
C	210 KB
D	220 KB
E	50 KB
F	16 KB
G	100 KB

(ii) Release the allocation to processes B, E, F and G. (2 marks)

(b) Figure 5.1 shows the pseudocode of a process with three shared variables and two threads.

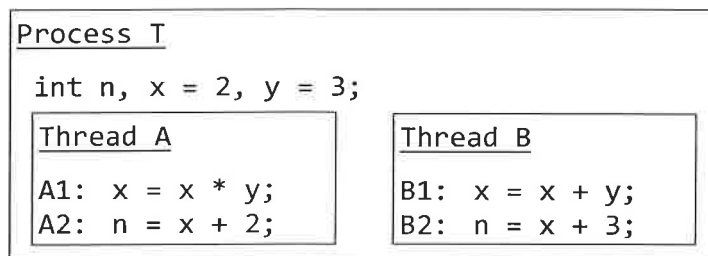


Figure 5.1

(i) Briefly explain race condition. (2 marks)

(ii) Show that race condition can occur in Process T. You may use the line number (A1, A2, etc.) in your explanation. You are not required to consider all the race condition scenarios; it is sufficient to include the minimum number of scenarios that can show the occurrence of race condition in your answer. (6 marks)

(iii) Assume that either thread can be selected to execute first by the scheduler. Write the pseudocode for both threads, as in Figure 5.1, based on the following requirements: (6 marks)

- Whenever the process T starts, Thread A is always performing the computations of x and n in its critical section before Thread B does the same.
- If Thread B is selected to execute first, it will be blocked from entering its critical section until Thread A exits from its critical section.
- A thread must acknowledge the other thread when it exits from its critical section so that the other thread can enter to the critical section.

UCCD2103 OPERATING SYSTEMS**Q5. (b) (Continued)**

- Use semaphores to control the entrance to the critical section in each thread and the semaphores must be initialised with the appropriate values.
- (c) Briefly explain how the direct prevention method prevents the occurrence of deadlock. (2 marks)
- (d) What are the **TWO (2)** problems in enforcing mutual exclusion with interrupt disabling? (2 marks)
- [Total : 25 marks]

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APPENDIX A:

Linux Completely Fair Scheduler (CFS):

1. Tasks' time slice allocation:

$$f_i(t) = \frac{w_i}{\sum w_{rbtree}} \times p$$

$f_i(t)$ = time slice received by task i at time t ,
 w_i = weight of task i

$\sum w_{rbtree}$ = Total weight of tasks in red-black tree

p = period

2. Virtual runtime of task i in n -th iteration:

$$v_i(n) = v_i(n-1) + \left(\frac{\Delta e}{w_i} \right) \times w_{nice0}$$

$v_i(n-1)$ = virtual runtime of task i in previous iteration

Δe = elapsed time task i spent utilizing the processor during iteration n

w_i = weight of task i

w_{nice0} = weight assigned to nice 0 task = 1024

3. Mapping of nice value to weight:

```
static const int prio_to_weight[40] = {
```

```
/* -20 */      88761,      71755,      56483,      46273,      36291,
/* -15 */      29154,      23254,      18705,      14949,      11916,
/* -10 */      9548,       7620,       6100,       4904,       3906,
/* -5  */      3121,       2501,       1991,       1586,       1277,
/* 0   */      1024,       820,        655,        526,        423,
/* 5   */       335,       272,        215,        172,        137,
/* 10  */       110,       87,         70,         56,         45,
/* 15  */        36,       29,         23,         18,         15,
}
```

